

Multistate models

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ms-Markov

Common assumptions in survival analysis

1. Subjects are **either** “healthy” **or** “diseased”, with no intermediate state.
2. The disease is **irreversible**, or requires intervention to be cured.
3. The disease is **accurately** diagnosed.
4. The time of disease incidence is known **exactly**.

These assumptions are true for **death** and many **chronic diseases**.

A question of definition:

—we use occurrence of **recording of** a given disease

Accurate event dates?

- ▶ **Accurate dates of transition:**
complete observation of the process
all times and types of transitions between states are known
- ▶ **Panel data**
states known at certain times for each person
not necessarily the same times for different persons
- ▶ Panel data is analyzed with the `msm` package
this representation not treated further here

Markov models for multistate processes

- ▶ transition **probability** between two states ($A \rightarrow B$) depends only on current state through $t - s$:

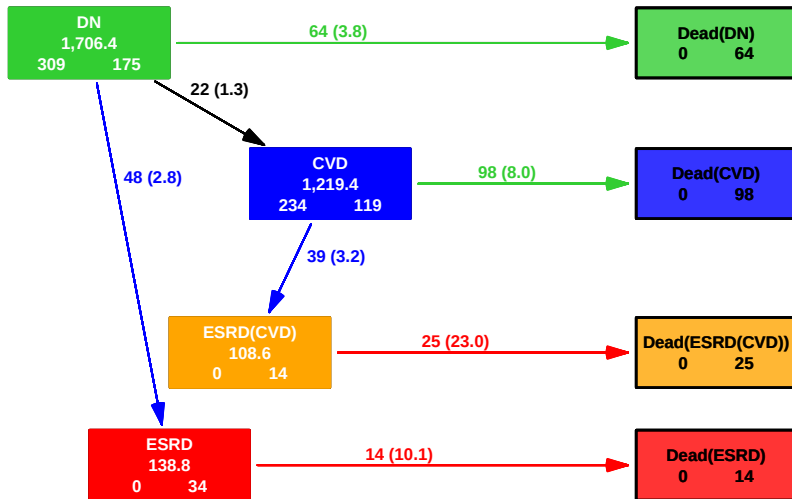
$$P \{ \text{state } B \text{ at } t \mid \text{state } A \text{ at } s \} = f(t - s)$$

- ▶ — this is the **Markov** property
- ▶ $\Rightarrow$ transition **rates** are constant over time
- ▶ (time-fixed) covariates may influence transition rates
- ▶ the formal Markov property is **very** restrictive
- ▶ in the clinical literature, the term “Markov model” is often used about any type of multistate model

Components of a multistate (Markov) model

- ▶ Define the disease states
- ▶ Define which transitions between states are allowed
- ▶ Select covariates influencing transition rates (may be different between transitions)
- ▶ Not a trivial task — do we want *e.g.*
 - ▶ cause of death (CVD, Cancer, Other)
 - ▶ disease status at death (prev.CVD, prev.Can, neither)
- ▶ Types of parameters to target:
 - ▶ **Transition rates** between states
 - ▶ **Probability** of **state occupancy**
 - ▶ Expected **sojourn time** in a state

A more complicated multistate model



Likelihood for a multistate model

- ▶ The likelihood of the model depends on the probability of being in state B at time t_1 , given that you were in state A at time t_0 .
- ▶ Assume transition rates constant in small time intervals, $\lambda^{A \rightarrow B}$
- ▶ $\Rightarrow$ each interval for a person contributes term(s) to the log-likelihood
- ▶ one term for each possible transition to a subsequent state
- ▶ the total log-likelihood for person p in state A during interval i is a sum of these terms: $\ell_p = \sum_{i,B} d_{pi} (\log(\lambda_{pi}^{A \rightarrow B}) - \lambda_{pi}^{A \rightarrow B} y_{pi})$
- ▶ $\Rightarrow$ each term has the form of the likelihood for a Poisson variate d with mean λy : $d \log(\lambda) - \lambda y$

Likelihood for a multistate model

- ▶ each term has the form of the likelihood for a Poisson variate d with mean λy
- ▶ terms are **not** independent, but the total likelihood is a **product**; hence of the same form as the likelihood from independent (Poisson) variates
- ▶ but terms from intervals within one person are neither Poisson nor independent

Realms of multistate modeling

- ▶ transition rates (intensities) — dimension time^{-1}
this is the scale of observation, (d, y) (complete data)
- ▶ state probabilities — dimensionless, time^0
 integral of intensities w.r.t. to time
- ▶ sojourn times — dimension time^1
 integral of state probabilities w.r.t. to time

Classes of multistate models

- ▶ Markov model: transition between states depends **only** on current state $\Rightarrow$ transition rates are constant
time-homogeneous Markov model
- ▶ If transition rates depend on the **same timescale** only we have a time-inhomogeneous Markov model
- ▶ If transition rates depend on the time since entry to the current state we have a semi-Markov model
- ▶ If transition rates depend on several timescales we have a general multistate model (there is no formal name for this)

...it is common-place in the literature to use the term “Markov model” for any type of multistate model.

Computing state probabilities from intensities in multistate models

- ▶ time-homogeneous Markov model:
closed-form formulae exist
- ▶ time-**inhomogeneous** Markov model:
closed-form formulae exist (a bit more complicated)
- ▶ **semi**-Markov model:
no closed form formulae exist
- ▶ general multistate model:
no closed form formulae exist

No formulae means that any inference on state probabilities and sojourn times must be based on (micro)**simulation** from the model.

Multistate models with Lexis

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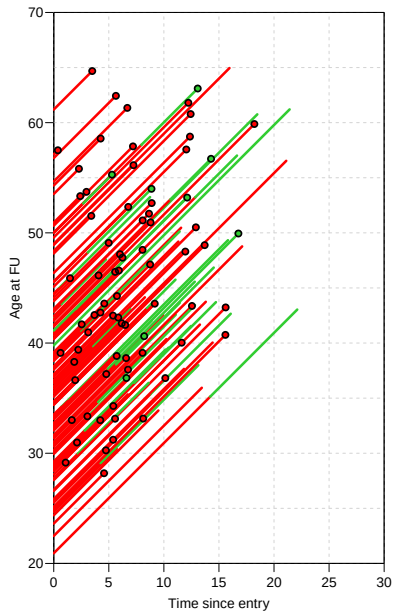
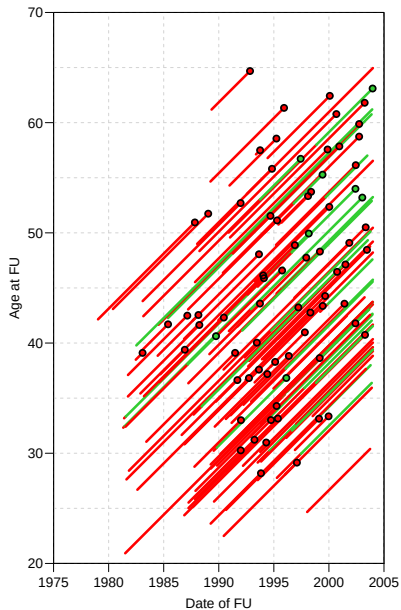
ms-Lexis

Example: Renal failure data from Steno

Hovind P, Tarnow L, Rossing P, Carstensen B, and Parving H-H: Improved survival in patients obtaining remission of nephrotic range albuminuria in diabetic nephropathy. *Kidney Int.*, 66(3):1180–1186, 2004.

- ▶ Endpoint of interest: Death or end stage renal disease (ESRD), i.e. dialysis or kidney transplant.
- ▶ 96 patients entering at nephrotic range albuminuria (NRA), i.e. $\text{U-alb} > 300\text{mg/day}$.
- ▶ Is remission from this condition (i.e return to $\text{U-alb} < 300\text{mg/day}$) predictive of the prognosis?

		Remission	
		Total	
			Yes No
No. patients		125	32 93
No. events		77	8 69
Follow-up time (years)		1084.7	259.9 824.8
Cox-model:			
Timescale:	Time since nephrotic range albuminuria (NRA)		
Entry:	2.5 years of GFR-measurements after NRA		
Outcome:	ESRD or Death		
Estimates:	RR	95% c.i.	p
Fixed covariates:			
Sex (F vs. M):	0.92	(0.53,1.57)	0.740
Age at NRA (per 10 years):	1.42	(1.08,1.87)	0.011
Time-dependent covariate:			
Obtained remission:	0.28	(0.13,0.59)	0.001



Features of the analysis

- ▶ Remission is included as a time-dependent variable.
- ▶ Age at entry is included as a fixed variable.

```
renal[1:5,]  
id      dob      doe      dor      dox event  
17 1967.944 1996.013      NA 1997.094      2  
26 1959.306 1989.535 1989.814 1996.136      1  
27 1962.014 1987.846      NA 1993.239      3  
33 1950.747 1995.243 1995.717 2003.993      0  
42 1961.296 1987.884 1996.650 2003.955      0
```

Note patient 26, 33 and 42 obtain remission.

```

> Lr <- Lexis(entry = list(per = doe,
+                           age = doe - dob,
+                           tfi = 0),
+             exit = list(per = dox),
+             exit.status = event > 0,
+             states = c("NRA", "ESRD"),
+             data = renal)
> summary(Lr)

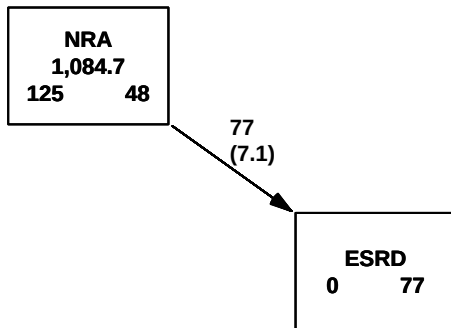
```

Transitions:

To

From	NRA	ESRD	Records:	Events:	Risk time:	Persons:
NRA	48	77	125	77	1084.67	125

```
> boxes(Lr, boxpos = list(x = c(25, 75),  
+                          y = c(75, 25)),  
+       scale.R = 100, show.BE = TRUE )
```



Cutting follow-up at remission: cutLexis

```
> Lc <- cutLexis(Lr, cut = Lr$dor,  
+               timescale = "per",  
+               new.state = "Rem")  
> summary(Lc)
```

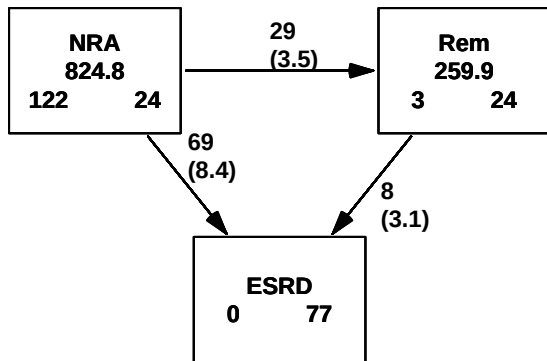
Transitions:

To

From	NRA	Rem	ESRD	Records:	Events:	Risk time:	Persons:
NRA	24	29	69	122	98	824.77	122
Rem	0	24	8	32	8	259.90	32
Sum	24	53	77	154	106	1084.67	125

Showing states and FU: boxes.Lexis

```
> boxes(Lc, boxpos = list(x = c(15, 85, 50),  
+                          y = c(85, 85, 20)),  
+       scale.R = 100, show.BE = TRUE)
```



Cutting follow up at events: cutLexis

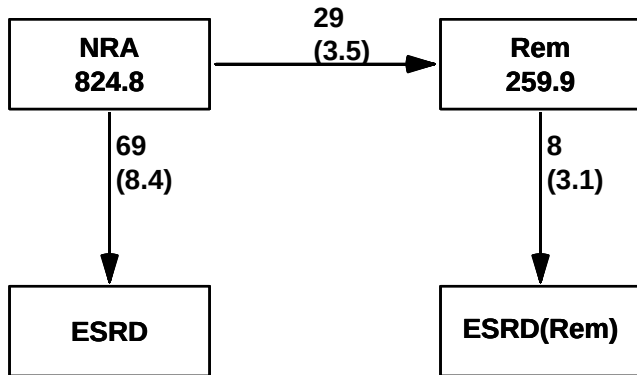
```
> Lc <- cutLexis( Lr, cut = Lr$dor,  
+               timescale = "per",  
+               new.state = "Rem",  
+               split.states = TRUE)  
> summary( Lc )
```

Transitions:

	To							
From	NRA	Rem	ESRD	ESRD(Rem)	Records:	Events:	Risk time:	Persons:
NRA	24	29	69	0	122	98	824.77	122
Rem	0	24	0	8	32	8	259.90	32
Sum	24	53	69	8	154	106	1084.67	125

Showing states and FU: boxes.Lexis

```
> boxes(Lc, boxpos = list(x = c(15, 85, 15, 85),  
+                          y = c(85, 85, 20, 20)),  
+       scale.R = 100)
```



Likelihood for a general MS-model

- ▶ Product of likelihoods for each transition
— each one as for a survival model
- ▶ **Risk time** is the risk time in the “From” state
- ▶ **Events** are transitions to the “To” state
- ▶ All other transitions out of “From” are treated as **censorings**
- ▶ Possible to fit models to **Lexis** objects
 - ▶ separately for each transition
 - ▶ jointly for transitions **from different** states **to the same** state
 - ▶ jointly for transitions **from the same** state **to different** states: but
don't!

Calculating state probabilities

P {Remission **before** time t }

$$= \int_0^t \lambda(u) \exp \left(- \int_0^u \lambda(s) + \mu_{\text{NRA}}(s) \, ds \right) \, du$$

P {Being in remission **at** time t }

$$= \int_0^t \lambda(u) \exp \left(- \int_0^u \lambda(s) + \mu_{\text{NRA}}(s) \, ds \right) \times \\ \exp \left(- \int_u^t \mu_{\text{rem}}(s) \, ds \right) \, du$$

Note μ_{rem} could also depend on u , time since obtained remission.

Sketch of programming, assuming that λ (`lambda`), μ_{NRA} (`mu.nra`) and μ_{rem} (`mu.rem`) are known at any age (stored in vectors)

```
c.rem      <- cumsum(lambda)
c.mort.nra <- cumsum(mu.nra)
c.mort.rem <- cumsum(mu.rem)
pr1 <- cumsum(lambda * exp(-(c.rem + c.mort.nra)))

intgr(t,s) <-
function(t,s){
  lambda[s] * exp(-(c.rem[s] + c.mort.nra[s])) *
    exp(-(c.mort.rem[t] - c.mort.rem[s]))
}
for(t in 1:100) p2[t] <- sum(intgr(t,1:t))
```

If μ_{rem} also depends on time since remission, then `c.mort.rem` should have an extra argument—technically very complicated

Analytical approach is not recommended — simulation is simpler

Prediction in multistate models: simLexis and renal failure

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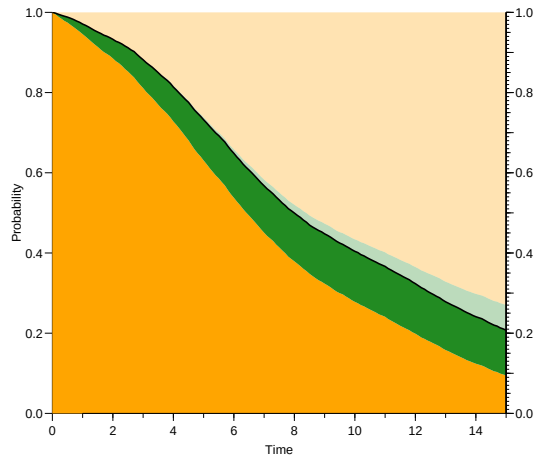
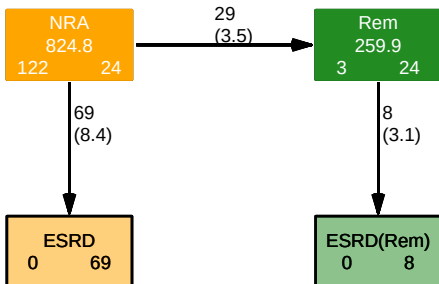
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simRenal



Modeling rates in a multistate model

Each transition modeled by a model for rates

(Cox-model or Poisson-model for split data: `glm` or `gam`)

Requires that follow-up is split in small intervals:

```
> sLc <- splitLexis(Lc, "tfi", breaks = seq(0, 30, 1/12))  
> summary(sLc, t = T)
```

Transitions:

		To							
From	NRA	Rem	ESRD	ESRD(Rem)	Records:	Events:	Risk time:	Persons:	
NRA	9854	29	69	0	9952	98	824.77	122	
Rem	0	3139	0	8	3147	8	259.90	32	
Sum	9854	3168	69	8	13099	106	1084.67	125	

Timescales:

```
per age tfi  
"" "" ""
```

Modeling rates in a multistate model

```
> # Remission-rate
> mr <- gam(cbind(lex.Xst == "Rem", lex.dur)
+          ~ s(tfi, k = 10) + sex,
+          family = poisreg,
+          data = subset(sLc, lex.Cst == "NRA"))
> #
> mr <- gamLexis( sLc,
+               ~ s(tfi, k = 10) + sex,
+               to = "Rem" )
```

mgcv::gam Poisson analysis of Lexis object sLc with log link:
Rates for the transition:
NRA->Rem

... using the Lexis properties

```
> # ESRD-rates
> mx <- gam(cbind(lex.Xst %in% c("ESRD", "ESRD(Rem)"), lex.dur)
+           ~ s(tfi, k = 10) + sex + I((doe - dob - 40) / 10) +
+           I(lex.Cst == "Rem"),
+           family = poisreg,
+           data = subset(sLc, lex.Cst %in% c("NRA", "Rem")))
> #
> mx <- gamLexis( sLc,
+               ~ s(tfi, k = 10) + sex + I((doe - dob - 40) / 10) +
+               lex.Cst )
```

mgcv::gam Poisson analysis of Lexis object sLc with log link:

Rates for transitions:

NRA->ESRD

Rem->ESRD(Rem)

Default is to model all transitions **to** absorbing states

State probabilities

How do we get from rate-models (and origin) to state probabilities:

1 Analytic calculations:

- ▶ immensely complicated formulae
- ▶ computationally fast (once implemented)
- ▶ difficult to generalize

2 Simulation of persons' histories

- ▶ conceptually simple
- ▶ computationally not quite simple
- ▶ easy to generalize
- ▶ hard to get confidence intervals (bootstrap)

Simulation of a survival time

- ▶ For a rate function $\lambda(t)$, $\Lambda(t) = \int_0^t \lambda(s) \, ds$:

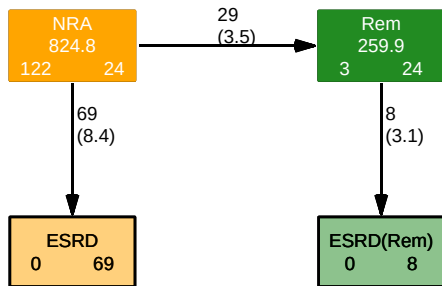
$$S(t) = \exp(-\Lambda(t))$$

- ▶ Simulate a survival probability $u \in [0, 1]$:

$$u = S(t) \quad \Leftrightarrow \quad \Lambda(t) = -\log(u)$$

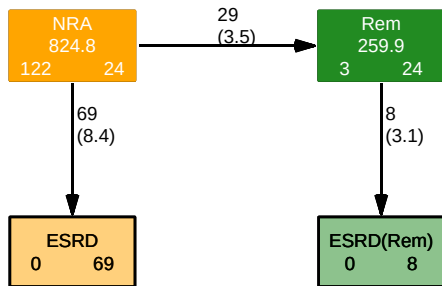
- ▶ Knowledge of $\Lambda(t)$ makes it easy to find a survival time — essentially just linear interpolation.

Simulation in a multistate model



- ▶ Simulate a “survival time” for each transition **out** of a state.
- ▶ The smallest of these is the transition time.
- ▶ Choose the corresponding transition type as transition.

Transition objects are fitted coxph/glm/gam models



```
> Tr <- list("NRA" = list("ESRD"      = mx,  
+                          "Rem"      = mr),  
+           "Rem" = list("ESRD(Rem)" = mx))
```

simLexis

Input required:

- ▶ A `Lexis` object with the initial state of the persons to be simulated.
(`lex.dur` and `lex.Xst` will be ignored—they are outcomes to be simulated)
- ▶ A transition object with the estimated Poisson models collected in a list of lists.

Output produced:

- ▶ A `Lexis` object with simulated event histories for many persons

Using simLexis I

Put one record a new Lexis object (`init`, say). representing a person with the desired covariate values.

Must have same structure as the one used for estimation — time scales must be initiated even if not used in models

```
> init <- sLc[NULL, c(timeScales(sLc), "lex.Cst")]
> init[1,"per"] <- 1994
> init[1,"age"] <- 40
> init[1,"tfi"] <- 0
> init[1,"lex.Cst"] <- "NRA"
> init[1,"sex"] <- "M"
> init[1,"dob"] <- 1954
> init[1,"doe"] <- 1994
> init
```

per	age	tfi	lex.Cst	sex	dob	doe
1994	40	0	NRA	M	1954	1994

Using simLexis II

```
> system.time( sim1 <- simLexis(Tr, init, N = 10000, t.range = 15.1) )
```

```
bruger    system forløbet  
20.55      1.50      22.28
```

```
> summary(sim1)
```

Transitions:

To

From	NRA	Rem	ESRD	ESRD(Rem)	Records:	Events:	Risk time:	Persons:
NRA	928	1759	7313	0	10000	9072	72462.56	10000
Rem	0	1119	0	640	1759	640	14882.74	1759
Sum	928	2878	7313	640	11759	9712	87345.29	10000

This is a simulated cohort of 10,000 persons with NRA, aged 40 in 1994.

Using a simulated Lexis object — pState I

```
> NN <- nState(sim1, at = seq(0, 15, 0.1),  
+             from = 0,  
+             time.scale = "tfi")  
> head(NN)
```

	State			
when	NRA	Rem	ESRD	ESRD(Rem)
0	10000	0	0	0
0.1	9951	30	19	0
0.2	9891	61	48	0
0.3	9841	85	74	0
0.4	9786	117	97	0
0.5	9746	135	119	0

```
> sP1 <- pState(NN, perm = c(1, 2, 4, 3))  
> head(sP1, 3)
```

Using a simulated Lexis object — pState II

```
      State
when      NRA      Rem ESRD(Rem) ESRD
  0      1.0000 1.0000      1.0000    1
  0.1 0.9951 0.9981      0.9981    1
  0.2 0.9891 0.9952      0.9952    1
```

```
> tail(sP1, 3)
```

```
      State
when      NRA      Rem ESRD(Rem) ESRD
14.8 0.0994 0.2134      0.2750    1
14.9 0.0972 0.2108      0.2729    1
15   0.0948 0.2075      0.2705    1
```


Using a simulated Lexis object — pState III

```
> par(mar = c(3, 3, 0.5, 2), mgp = c(3, 1, 0) / 1.6, las = 1)
> plot(sP1, col = clr[c(1, 2, 3, 4)], xlim = c(0,15) )
> lines(as.numeric(rownames(sP1)), sP1[,2], lwd = 2)
> axis(side = 1, at = 1:55, labels = NA, tcl = -0.3)
> axis(side = 4, at = 0:5 / 5)
> axis(side = 4, at = 0:10 / 10, labels = NA)
> axis(side = 4, at = 0:20 / 20, labels = NA, tcl = -0.3)
> axis(side = 4, at = 0:100/100, labels = NA, tcl = -0.2)
```

```
> sP2 <- pState(NN, perm = c(4, 2, 1, 3))
> head(sP2, 3)
```

	State				
when	ESRD(Rem)	Rem	NRA	ESRD	
0	0	0.0000	1.0000	1	
0.1	0	0.0030	0.9981	1	
0.2	0	0.0061	0.9952	1	

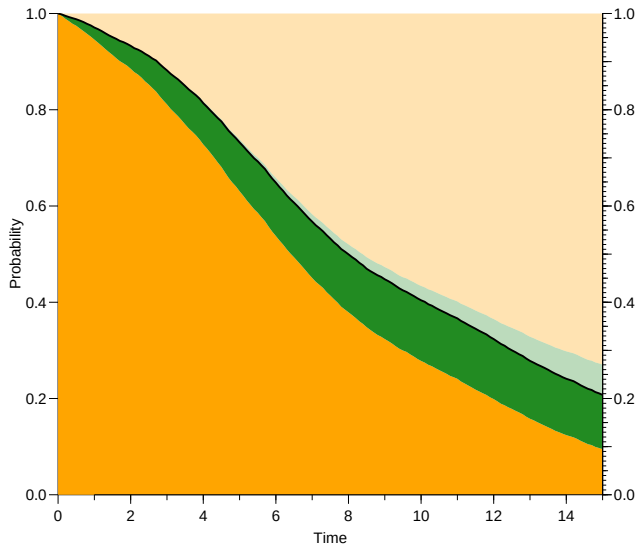
Using a simulated Lexis object — pState IV

```
> tail(sP2, 3)
```

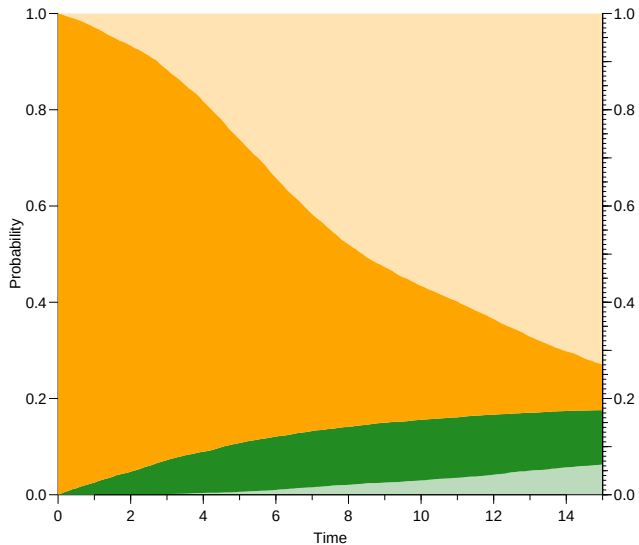
	State				
when	ESRD(Rem)	Rem	NRA	ESRD	
14.8	0.0616	0.1756	0.2750	1	
14.9	0.0621	0.1757	0.2729	1	
15	0.0630	0.1757	0.2705	1	

```
> par(mar = c(3, 3, 0.5, 2), mgp = c(3, 1, 0) / 1.6, las = 1)
> plot(sP2, col = clr[c(3, 2, 1, 4)])
> axis(side = 1, at = 1:55, labels = NA, tcl = -0.3)
> axis(side = 4, at = 0:5 / 5)
> axis(side = 4, at = 0:10 / 10, labels = NA)
> axis(side = 4, at = 0:20 / 20, labels = NA, tcl = -0.3)
> axis(side = 4, at = 0:100/100, labels = NA, tcl = -0.2)
```

Simulated probabilities



Simulated probabilities



How many persons should you simulate?

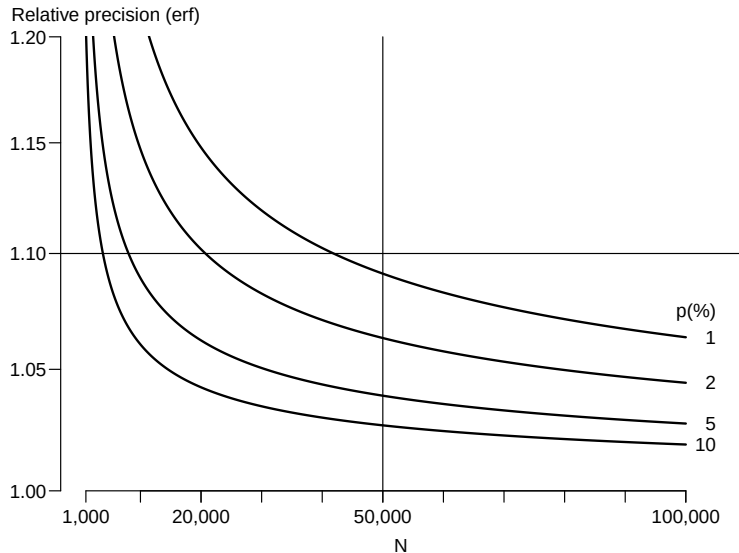
- ▶ All probabilities have the same denominator — the initial number of persons in the simulation, N , say.
- ▶ Thus, any probability will be of the form $p = x/N$
- ▶ For small probabilities we have that:

$$\text{s.e.}(\log(\hat{p})) = (1 - p) / \sqrt{Np(1 - p)}$$

- ▶ So c.i. has the form $p \div^{\times} \text{erf}$ where:

$$\text{erf} = \exp(1.96 \times (1 - p) / \sqrt{Np(1 - p)})$$

Precision of simulated probabilities



Using a simulated Lexis object — sohour time I

```
> summary(sim1)
```

Transitions:

To

From	NRA	Rem	ESRD	ESRD(Rem)	Records:	Events:	Risk time:	Persons:
NRA	928	1759	7313	0	10000	9072	72462.56	10000
Rem	0	1119	0	640	1759	640	14882.74	1759
Sum	928	2878	7313	640	11759	9712	87345.29	10000

```
> xtabs(lex.dur / 10000 ~ lex.Cst, data = sim1) /> addmargins()
```

lex.Cst

NRA	Rem	ESRD	ESRD(Rem)	Sum
7.246256	1.488274	0.000000	0.000000	8.734529

During the first 15 years after entry, the expected time in remission is 1.5 years out of an expected lifetime of 8.7 years, for a 40 year old man

Multistate model overview

- ▶ Clarifies the relevant states and transitions are
- ▶ Allows proper estimation of transition rates
- ▶ — and relationships between them
- ▶ Separate model for each transition
- ▶ The usual survival methodology to compute probabilities breaks down
- ▶ Simulation allows estimation of cumulative probabilities:
 - ▶ Estimate transition rates (as usual)
 - ▶ Simulate probabilities (**not** quite as usual)